1. Which of the following are required for sharing a data set? 1 point

- ◯ A code book describing each variable and its values
- ◯ A tidy data set
- ◉ All of these options
- ◯ The raw data

2. Which of the following should be included in data tidying recipes? 1 point

- ◯ Units of variables
- ◯ Power calculations
- ◉ Parameter values for all functions
- ◯ Preprocessed data

3. What is the central dogma of statistics? 1 point

- ◯ That increased power comes with increased sample sizes
- ◯ Using Bayes rule to calculate probabilities we care about
- ◯ Using measurements on a population to infer knowledge about a sample
- ◉ Using measurements on a probabilistically selected sample to infer knowledge about a population

4. Which of the following are types of variability in all genomic data? 1 point

- ◯ Phenotypic variability
- ◉ Genetic drift
- ◯ Variation from changing technology
- ◯ Variability due to dropout

5. Which of the following will increase power in a statistical analysis? 1 point

- ◯ Increasing sample size
- ◉ Using a new technology
- ◯ Adjusting for confounders
- ◯ Increasing measurement variation

6. If 100 p-values are calculated on a data set with no signal, how many p-values would we expect to be less than 0.05 on average? 1 point

- ◯ 0
- ◯ 50
- ◉ 5
- ◯ 0.05

7. If we report 500 results as significant out of 10,000 tests while controlling the family-wise error rate at 5%, about how many false positives do we expect?

1 point

- ○ 10
- ○ 20
- ○ 5
- ◉ 0

8. What is the most common confounder in genomics?

1 point

- ○ Sex
- ○ Genetic background
- ○ Population stratification
- ◉ Batch effects

9. Which of the following can be used to address potential confounders at the experimental design stage?

1 point

- ◉ Randomization
- ○ Measuring DNA instead of RNA
- ○ Multiple testing correction
- ○ Using linear models

10. Which of the following are benefits of making big data as small as possible as soon as possible?

1 point

- ○ Reducing the data will reduce the number of hypothesis tests
- ○ Reduced data will increase the power of statistical tests
- ◉ Interactive analysis can improve our ability to make discoveries
- ○ Smaller data sets will decrease false discovery rates